

Integral Relations among Totient Sections

Finite-level bases and integral relations

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ABSTRACT

We determine a basis and all integral relations for the sequences $n \mapsto \varphi(k^j n + r)$ with $0 \leq j \leq e$ and $0 \leq r < k^j$, where $k \geq 2$ and $e \geq 1$. The basis has $k^e + 1$ elements, and every other sequence is an explicit integer multiple of one of them. We prove independence by constructing nonsingular evaluation matrices and compare this argument with Martin's stronger affine theorem. Irrationality of the unreduced binary totient series remains open.

1 Introduction

Splitting the totient sequence into its even and odd subsequences, then splitting once more, gives seven sequences:

$$\varphi(n); \quad \varphi(2n), \varphi(2n+1); \quad \varphi(4n), \varphi(4n+1), \varphi(4n+2), \varphi(4n+3).$$

The identities

$$\varphi(4n) = 2\varphi(2n), \quad \varphi(4n+2) = \varphi(2n+1)$$

reduce this list to five sequences, which are linearly independent over $\mathbb{Q}$. Every rational linear identity among the seven therefore follows from these two substitutions. The substitutions also preserve integral coefficients, so they describe the integral relations as well.

The same description holds at every finite depth and in every base. For an integer $k \geq 2$, the *totient sections* are the sequences $n \mapsto \varphi(k^j n + r)$, where $j \geq 0$ and $0 \leq r < k^j$. Throughout, n ranges over $\mathbb{N} = \{0, 1, 2, \dots\}$ and $\varphi(0) = 0$. To describe relations we retain a separate index for each pair (j, r) , even when two pairs give the same sequence. For example, $\varphi(4n+2)$ and $\varphi(2n+1)$ are equal sequences but occupy different positions in the list above.

The sections form the k -kernel of the totient sequence. A sequence is k -regular when the $\mathbb{Z}$ -module generated by its k -kernel is finitely generated [1]. Coons proved that the totient sequence is not k -regular for any $k \geq 2$ [2, Thm. 3.2]. We describe the finitely generated modules obtained by stopping at a fixed level. The theorem below specifies a basis and expresses each remaining section as a scalar multiple of a basis element; these expressions give a test for every linear identity at that level. The level-zero family contains only $\varphi(n)$ and has dimension one.

**Authorship and AI use.* Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

Theorem 1.1 (A basis through each finite level). *Let $k \geq 2$ and $e \geq 1$ be integers, write $F_{j,r}^{(k)}(n) = \varphi(k^j n + r)$, and put*

$$V_{k,e} = \text{span}_{\mathbb{Q}} \{ F_{j,r}^{(k)} : 0 \leq j \leq e, 0 \leq r < k^j \}.$$

Then $\dim_{\mathbb{Q}} V_{k,e} = k^e + 1$, and

$$\mathcal{B}_{k,e} = \{F_{0,0}^{(k)}, F_{1,0}^{(k)}\} \cup \{F_{j,r}^{(k)} : 1 \leq j \leq e, 1 \leq r < k^j, k \nmid r\}$$

is a basis. Every omitted section reduces to a basis element by an explicit scalar: $F_{j,0}^{(k)} = k^{j-1} F_{1,0}^{(k)}$ for $j \geq 1$, and if $r = k^t u$ with $t = \max\{s : k^s \mid r\} \geq 1$ and $k \nmid u$, then

$$F_{j,r}^{(k)} = C_k(t, u) F_{j-t, u}^{(k)}, \quad C_k(t, u) = k^t \prod_{\substack{p \mid k \\ p \nmid u}} \left(1 - \frac{1}{p}\right).$$

The condition $k \nmid r$ allows residues with $\gcd(k, r) > 1$. For instance, $r = 1$ and $r = 2$ are both retained in base six, and the reductions involving them are

$$\varphi(36n + 6) = 2\varphi(6n + 1), \quad \varphi(36n + 12) = 4\varphi(6n + 2).$$

The multipliers differ because 2 divides $6n + 2$ but not $6n + 1$. The product formula also allows repeated prime factors of k , so there is no squarefreeness assumption on the base.

Martin’s 2006 theorem on simultaneous totient inequalities already implies independence for every finite family of pairwise nonproportional affine forms with positive integral slopes [3, Thm. 1]. Its conclusion is stronger than the independence used here. The point of the present description is the explicit reduction of the sections and the integral relation basis in Corollary 3.1. We also give a direct proof that constructs a nonsingular evaluation matrix. It uses the CRT–Dirichlet separation argument in Yazdani’s proof of Theorem 2, credited there to Shallit [5, pp. 652–653]. The step that needs care is to keep the progression for the selected affine form reduced while imposing prime divisors on all the others.

Section 2 proves the basis theorem. Section 3 starts with the six reductions in base six and derives the integral relation basis. The series $S = \sum_{n \geq 1} \varphi(n) 2^{-n}$ has unbounded coefficients, and its irrationality remains open (Erdős Problem #249 [4, p. 61]). The basis theorem supplies no irrationality conclusion for this value. The [long record’s residue classification](#) and [exponential-sum question](#) pursue the distinct arithmetic questions.

2 Proof of the basis theorem

Scalar reductions

There are two parts to the proof. The product formula reduces every omitted section to a retained one; independence then shows that no further reductions are possible. In the first part, the multiplier depends only on which prime divisors of k divide the argument. For a zero residue, kn and $k^j n$ have the same prime divisors when $n > 0$. Hence $\varphi(k^j n) = k^{j-1} \varphi(kn)$; both sides vanish at $n = 0$. For an omitted positive residue $r = k^t u$ with $t \geq 1$

and $k \nmid u$, we have $t < j$ because $0 < r < k^j$. Put $m = k^{j-t}n + u$. A prime $p \mid k$ divides m exactly when it divides u . The product formula for $\varphi(k^t m)/\varphi(m)$ therefore contributes k^t and one factor $1 - 1/p$ for each prime $p \mid k$ absent from u . The multiplier is an integer: its denominator is a product of distinct primes dividing k , and therefore divides k^t . This gives $C_k(t, u)$ and proves that the stated family spans $V_{k,e}$.

Affine independence

Independence follows from Martin's theorem. For any ordering of pairwise nonproportional affine forms and any $C > 0$, it gives a set of positive lower density on which each successive totient ratio exceeds C . In a proposed nonzero relation, place a nonzero coefficient c_1 first and take $C > \max(1, \sum_{i>1} |c_i|/|c_1|)$. The first term then dominates the sum of all remaining terms in absolute value, a contradiction [3, Thm. 1]. This also proves independence after any finite prefix is deleted.

We give a direct proof by constructing a square matrix of evaluations with nonzero determinant. Let $L_i(n) = a_i n + b_i$ for $1 \leq i \leq s$, where $a_i > 0$ and $a_i b_j - a_j b_i \neq 0$ for $i \neq j$. A shift of n makes all the b_i positive and leaves these cross determinants unchanged. Write $g_i = \gcd(a_i, b_i)$ and $A_i = L_i/g_i$. If $g_i > 1$, the values of L_i have a fixed divisor, so it is the primitive form A_i , not L_i , that we will make prime. We will choose a prime ℓ and points $n_1, \dots, n_s$ for which

$$E = (\varphi(L_j(n_i)))_{1 \leq i, j \leq s} \equiv \begin{pmatrix} \varphi(g_1) & 0 & \cdots & 0 \\ 0 & \varphi(g_2) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \varphi(g_s) \end{pmatrix} \pmod{\ell}, \quad \ell \nmid \prod_i \varphi(g_i). \quad (1)$$

A rational relation among the sequences would give a vector in the kernel of E . The displayed congruence makes its determinant nonzero, so no such relation can exist.

For row i , we require $L_j(n_i)$ to have a prime divisor $q_{ij} \equiv 1 \pmod{\ell}$ whenever $j \neq i$. Then $q_{ij} - 1$ divides $\varphi(L_j(n_i))$, giving the off-diagonal zeros in (1). For the diagonal entry, we require $A_i(n_i)$ to be a prime congruent to 2 modulo ℓ . Choose an odd ℓ dividing none of the slopes a_i , the products $g_i \varphi(g_i)$ or the nonzero cross determinants. For each fixed i , Dirichlet's theorem supplies distinct primes $q_{ij} \equiv 1 \pmod{\ell}$ ($j \neq i$) outside the same finite exceptional set. We impose

$$A_i(n) \equiv 2 \pmod{\ell}, \quad L_j(n) \equiv 0 \pmod{q_{ij}} \quad (j \neq i).$$

Each congruence is soluble because its slope is coprime to its modulus. CRT combines them into $n = n_0 + Ru$, where $R = \ell \prod_{j \neq i} q_{ij}$.

Dirichlet's theorem must now be applied to the values of A_i , rather than to n . These values form the progression

$$A_i(n_0 + Ru) = A_i(n_0) + (a_i/g_i)Ru.$$

It remains to check that its initial value and step are coprime. Primitivity excludes the prime factors of a_i/g_i , and the residue 2 excludes the odd prime ℓ . A prime q_{ij} cannot divide $A_i(n_0)$: it would then divide both $L_i(n_0)$ and $L_j(n_0)$, hence

$$a_i L_j(n_0) - a_j L_i(n_0) = a_i b_j - a_j b_i,$$

contrary to its choice. This accounts for every prime factor of the step. We may therefore choose $p = A_i(n_i)$ prime, as large as needed, and in particular with $p \nmid g_i$. Then

$$\varphi(L_i(n_i)) = \varphi(g_i)(p-1) \equiv \varphi(g_i) \not\equiv 0 \pmod{\ell},$$

while all the other entries of row i vanish modulo ℓ . We choose the rows separately, applying Dirichlet's theorem once more for each selected form A_i . Thus no simultaneous primality assertion is required. The construction proves (1) with every n_i beyond any prescribed threshold.

The two zero-residue sections

The affine-independence argument does not separate $\varphi(n)$ and $\varphi(kn)$, because their arguments are proportional. We use two different values of their ratio. On $n = km + 1$ the ratio is $\varphi(k)$, so a proposed relation contains only one zero-residue term after this restriction. The positive-residue sections become the forms

$$k^{j+1}m + k^j + r \quad (k \nmid r),$$

together with $km + 1$. These are pairwise nonproportional: equality of two intercept-to-slope ratios would give $r/k^j = r'/k^{j'}$, which for $j' > j$ forces $k \mid r'$. No positive-residue form is proportional to $km + 1$. Affine independence eliminates every positive-residue coefficient and leaves $A + B\varphi(k) = 0$, where A, B multiply $\varphi(n), \varphi(kn)$. We now evaluate the remaining relation at $n = k$ to obtain $A + Bk = 0$. Since $\varphi(k) < k$, the two equations force $A = B = 0$. Finally,

$$|\mathcal{B}_{k,e}| = 2 + \sum_{j=1}^e (k^j - k^{j-1}) = k^e + 1,$$

which completes the proof of Theorem 1.1. For independence on a tail, use a sufficiently large power of k in the last evaluation.

3 Integral relations

Once independence is known, a linear identity can be tested by substitution alone: reduce to the retained sections and compare their coefficients. Consider base six at level two, where the family has 43 indices. The 37 retained sections are $F_{0,0}, F_{1,0}$, the five $F_{1,r}$ ($1 \leq r \leq 5$), and the 30 sections $F_{2,r}$ with $1 \leq r < 36$ and $6 \nmid r$. The omitted sections satisfy

$$\begin{aligned} F_{2,0} &= 6F_{1,0}, & F_{2,6} &= 2F_{1,1}, & F_{2,12} &= 4F_{1,2}, \\ F_{2,18} &= 3F_{1,3}, & F_{2,24} &= 4F_{1,4}, & F_{2,30} &= 2F_{1,5}. \end{aligned}$$

Thus an integral combination of the 43 sections vanishes precisely when these six substitutions leave all 37 retained coefficients zero. To recover the relation itself, keep track of what each substitution removes. Let $E_{j,r}$ denote the generator at index (j, r) in the free abelian group on all section indices. The substitution $F_{2,12} = 4F_{1,2}$ corresponds to the decomposition

$$x_{1,2}E_{1,2} + x_{2,12}E_{2,12} = (x_{1,2} + 4x_{2,12})E_{1,2} + x_{2,12}(E_{2,12} - 4E_{1,2}).$$

The first coefficient tests whether the retained component vanishes; the second records the multiple of the relation removed. Doing this at all six omitted indices gives every integral relation, since the retained sections are independent. At base two and level two the relations are $E_{2,0} - 2E_{1,0}$ and $E_{2,2} - E_{1,1}$. The following corollary expresses the same calculation at arbitrary base and level.

Corollary 3.1 (Integral coordinates and all integral relations). *Let $k \geq 2$ and $e \geq 1$. The retained family is a $\mathbb{Z}$ -basis of the module generated by the sections through level e . Index the sections by their level and residue, retaining distinct indices even when they define equal sequences. For each omitted index i , write the scalar reduction as $F_i = a_i F_{j(i)}$, where $j(i)$ is retained and a_i is a nonnegative integer. In the free abelian group with one generator E_i for each of these indices, the vectors*

$$R_i = E_i - a_i E_{j(i)} \quad (i \text{ omitted})$$

form a $\mathbb{Z}$ -basis of the kernel of evaluation $E_i \mapsto F_i$. In particular, every integral relation has a unique integral expression in these elementary relations, and their rank is $\sum_{j=1}^{e-1} k^j$.

Proof. Since $\prod_{p|k, p \nmid u} p$ divides k , every scalar $C_k(t, u)$ is an integer. The retained family therefore spans the section module over $\mathbb{Z}$, and its rational independence makes it a $\mathbb{Z}$ -basis.

For any vector $x = \sum_i x_i E_i$, substitute $E_i = R_i + a_i E_{j(i)}$ at each omitted index and collect terms:

$$x = \sum_{j \text{ retained}} \left(x_j + \sum_{\substack{i \text{ omitted} \\ j(i)=j}} a_i x_i \right) E_j + \sum_{i \text{ omitted}} x_i R_i.$$

Evaluation kills the second sum. Independence of the retained sections therefore says that x is a relation exactly when every coefficient in the first sum is zero. In that case the second sum is its expression in the R_i . This expression is unique: the coefficient at omitted index i in $\sum_h c_h R_h$ is c_i itself. The unit coefficient at that index is also why no division is needed. Counting omitted indices gives $(1 + k + \dots + k^e) - (k^e + 1) = \sum_{j=1}^{e-1} k^j$. $\square$

The assertion concerns the module generated by the sections. It need not contain every integer-valued sequence in its rational span: for $k = 2$ and $e \geq 2$, the sequence $\varphi(4n + 3)/2$ is integer-valued and has coefficient $1/2$ in the retained basis. Thus this module is not saturated in the group of integer-valued sequences.

Verification and further consequences

The verification concordance lists the formal proofs by statement. *Lean* links to the supporting declarations; a dagger identifies a proof that assumes a named input. *Comparator* links to a recorded kernel check against a separately written statement; *pending* means that this comparison has not been recorded. The [verification record](#) gives the precise correspondence, dependencies and reproducible checks. A row with only a record link has no complete formal proof recorded. The [long record's verification section](#) locates the

formal statements and the reproducible base-six calculation. Its [further consequences for the sections](#) include periodic coefficient relations and section operators. These results, and the separate rational-series criteria, are not needed for the basis and integral relations proved here.

References

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- [5] S. Yazdani, *Multiplicative functions and k -automatic sequences*, J. Théor. Nombres Bordeaux **13** (2001), no. 2, 651–658, https://www.numdam.org/item/JTNB_2001__13_2_651_0/. Theorem 2 and its proof, pp. 652–653; Corollary 4, p. 654.

Verification concordance

Statement	Formal proof	Statement	Formal proof
Theorem 1.1	Lean · Comparator	Corollary 3.1	Lean · Comparator